Journal of Clinical Child Psychology
2002, Vol. 31, No. 2, 252–262

Comparing Measures of Childhood Loneliness: Internal Consistency and Confirmatory Factor Analysis

Luc Goossens and Wim Beyers
Katholieke Universiteit Leuven

Compared 6 self-rating measures of loneliness and associated phenomena, designed for use with elementary-school children. Three samples of children in Grades 5 and 6 (total N = 758) completed various combinations of these instruments. In terms of internal consistency, the Children's Loneliness Scale (CLS) and the peer-related loneliness subscale of the Loneliness and Aloneness Scale for Children and Adolescents (LACA) were most reliable. Substantial support was obtained for the convergent validity of the measures. Confirmatory factor analysis (CFA) on 2 samples and exploratory factor analysis on a third sample indicated that the 13 scales and derivative subscales of the 6 measures tapped 4 different but interrelated latent constructs: peer-related loneliness, family-related loneliness (in some cases restricted to parent-related loneliness), negative attitude toward being alone, and positive attitude toward being alone. Recommendations are offered for conditions under which each scale may be useful. Limitations of this study are discussed and suggestions for future research are offered.

Loneliness, defined as a negative emotional response to a discrepancy between desired and achieved levels of social contact (Peplau & Perlman, 1982), is a problem that affects people of all ages, including elementary-school children (Rotenberg, 1999). In this particular age group, prolonged loneliness has been shown to lead to internalizing problems such as depression (Boivin, Hymel, & Bukowski, 1995). Such a link could be explained by the interpersonal theory of depression, which basically holds that the rich and reciprocal interactions between depressed people and the key people in their lives contribute to the emergence, consolidation, and maintenance of their depressive symptoms (Coyne, Burchill, & Stiles, 1991). Recent research on adults suggests that loneliness mediates depression in that a key component of loneliness, that is, lack of pleasurable engagement, may be a prominent feature of chronic depressive states (e.g., chronic major depression). A secondary component of loneliness, that is, painful disconnection from others, may be characteristic of more acute states (e.g., acute episodes of major depression; Joiner, Catanzaro, Rudd, & Rajab, 1999). To the extent that these findings also hold for elementary-school children, they provide important clues for prevention and intervention efforts at this early age.

Both the identification of lonely children and the evaluation of intervention efforts designed to curb the negative sequelae of loneliness depend, to a large extent, on adequate measurement of the phenomenon at hand. To make an informed choice among such instruments, researchers need an overview of the range of childhood loneliness measures currently available and detailed information on their reliability and validity. Thus, this series of studies compared the internal consistency of a broad range of childhood loneliness measures and checked their convergence by means of confirmatory factor analysis (CFA).

Loneliness Measures Currently Available

A great number of scales are available to researchers who want to measure children's loneliness. At the heart of this diversity lies a double ambition. Researchers have (a) tried to tap various types of loneliness as experienced within different relationships and (b) examined related constructs to place children's feelings of loneliness in a somewhat broader perspective. Relationships with parents and peers have received ample attention (Bullock, 1993), and attitude toward being alone was selected as a pivotal construct by several researchers.

We are greatly indebted to three anonymous reviewers for their insightful comments on earlier drafts of this article. Special thanks go to Esther Beckers, An Callewaert, and Sofie Derez for their assistance in data collection; to Rob Stroobants for his statistical assistance; and to Alfons Marcoen, Griet Van Roosmalen, and Veerle Germeijs for their help in translating the measures into Dutch.

Covariance matrices for all three samples are available from Wim Beyers.

Requests for reprints should be sent to Luc Goossens, Katholieke Universiteit Leuven, Center for Developmental Psychology, Tiensestraat 102, B–3000 Leuven, Belgium. E-mail: Luc.Goossens@psy.kuleuven.ac.be

Specifically, children's attitude toward being alone is thought to moderate the relation between solitude and the experience of loneliness. A child's affinity for aloneness could act as a buffer against feeling lonely when alone. Children's aversion to aloneness, by contrast, might increase their vulnerability to feeling lonely when alone (Youngblade, Berlin, & Belsky, 1999).

Six loneliness measures are currently available: the Children's Loneliness Scale (CLS; Asher, Hymel, & Renshaw, 1984; Asher & Wheeler, 1985), the Loneliness and Aloneness Scale for Children and Adolescents (LACA; formerly known as the Louvain Loneliness Scale for Children and Adolescents; Marcoen, Goossens, & Caes, 1987), the Children's Multidimensional Loneliness Scale (CMLS; DeBiase, 1992), the Relational Provisions Loneliness Questionnaire (RPLQ; Hayden-Thomson, 1989), the Children's Emotional Loneliness Scale (CELS; Heinlein & Spinner, 1986), and the Ability to Be Alone Questionnaire (ABAQ; Youngblade et al., 1999). The 13 scales and derivative subscales of these instruments are indicators of one of the four concepts that have captured researchers' interest: peer-related loneliness, parent-related loneliness, aversion to aloneness, and affinity for aloneness.

Peer-related loneliness is assessed by six subscales: the CLS, the two peer-related loneliness scales of the RPLQ, the peer-related loneliness subscale found in the LACA, and the peer-related and global loneliness subscales of the CMLS. All items of the latter subscale refer to the generalized other who, in view of the particular structure of the social world of late childhood, is most likely to be a peer.

Parent-related (or *family-related*) *loneliness* is assessed by five subscales: the two family-related loneliness scales of the RPLQ, the parent-related loneliness subscales found in the LACA and the CMLS, respectively, and the CELS. The latter scale actually has the most varied item content of all childhood loneliness measures currently available; some of the items do refer to children's bonds with family members, whereas other items refer to unidentified adults and to the generalized other.

Aversion to being alone is assessed by two subscales: the Aversion to Being Alone subscale as found in both the LACA (Marcoen et al., 1987) and the ABAQ (Youngblade et al., 1999). *Affinity for being alone*, finally, is also assessed by two subscales: the Affinity for Being Alone subscale of the LACA and the Ability to Be Alone subscale of the ABAQ.

A Four-Construct Model of Loneliness and Related Phenomena

Some authors (e.g., Marcoen & Goossens, 1993) have claimed that the four constructs, when taken together, constitute a heuristic model of loneliness and associated phenomena in childhood and adolescence. The two types of loneliness refer to the two major types of relationships in that part of the life span, that is, relationships with parents and with friends. The two attitudes toward being alone represent two basic types of generalized reactions to being on one's own that emerge during that same period. Researchers with an active interest in the early phases of life, therefore, ought to measure all four constructs or to keep in mind how both loneliness and reactions to being alone are conditioned by their developmental context and may change with advancing age.

Parents, for instance, are gradually being replaced by friends as primary attachment figures, but because they continue to be important to their children throughout adolescence, parent-related loneliness actually increases, particularly in mid- to late adolescence. Peer-related loneliness, by contrast, decreases from early adolescence onward, when an important shift toward greater intimacy comes to characterize relationships with peers (Marcoen et al., 1987). Children typically have a mostly negative attitude toward being alone, whereas adolescents gradually become more introspective and reflective and come to ascribe conscious and deliberate functions to being alone (Larson, 1990). As a consequence, scores on the Negative Attitude to Aloneness scale tend to decrease during adolescence, whereas affinity for aloneness increases (Goossens & Marcoen, 1999). These average developmental trends hold for both boys and girls (Marcoen & Goossens, 1993).

Both the differential functions in development and the distinctive age trend for each construct imply that the four-construct model of loneliness and associated phenomena is not merely empirically derived but is firmly grounded in both developmental theory and empirical findings on children and adolescents. However, before researchers set out to check these supposed functions and hypothesized developmental trends, they have to address two basic psychometric questions. These questions simply are whether the four components of the model can be reliably measured and whether they can be actually distinguished from one another.

Internal Consistency

Research indicated that all measures of parent-related and peer-related loneliness, as found in the CLS (Crick & Ladd, 1993; Parker & Asher, 1993; Parkhurst & Asher, 1992), the CELS (Heinlein & Spinner, 1986), the RPLQ (Terrell-Deutsch, 1990), the CMLS (DeBiase, 1992), and the LACA (Goossens & Marcoen, 1999; Marcoen & Goossens, 1993; Terrell-Deutsch, 1990) were sufficiently reliable to be used for comparisons among groups (i.e., Cronbach's α reached .80 or above). The subscales on children's attitude to being alone, however, failed to reach that standard. Slightly lower estimates of internal consistency (in the .70s) were obtained for the two subscales of the LACA that probed children's attitude to being alone (Terrell-Deutsch, 1990) and for the subscales of the ABAQ (Youngblade et al., 1999).

CFA

A basic form of validity entails that multiple measures of the same construct exhibit a high degree of agreement (convergence) and that measures of different constructs can be clearly differentiated from one another (discrimination). Indications of this type of validity can be gained from an examination of the raw intercorrelations among childhood loneliness measures (see Terrell-Deutsch, 1999). A stronger validity test, which explicitly takes measurement error into account, is provided by CFA or latent structure analysis (Long, 1983). When the loneliness scales and derivative subscales are completed by a reasonably large sample of children, one may test statistically whether their pattern of intercorrelations as a whole is consistent with the four-construct model described earlier.

This model may be compared to alternative models in terms of their overall agreement with the data. Simpler or more complex models may provide a better fit to the observed pattern of intercorrelations or aspects thereof. The peer-related and parent-related loneliness measures, for instance, may not define two latent constructs, but rather a common construct (labeled "global loneliness"). Alternatively, three latent constructs (i.e., peer-related loneliness, parent-related loneliness, and loneliness related to the generalized other) may provide a more accurate description of the intercorrelations among these measures. In short, CFA or structural equation modeling, though still plagued by the fact that high correlations can partially reflect common method variance, allows researchers to empirically test current conceptualizations of childhood loneliness.

This Study

In this study, we re-examined the internal consistency of the six loneliness measures currently available. We expected that all the loneliness subscales would evidence a Cronbach's α of .80 or above, whereas the subscales that focus on children's attitude to being alone would fail to reach that commonly accepted benchmark. In addition, we used latent structure analysis to compare the four-factor model of childhood loneliness to a series of alternative models.

For practical reasons, however, we were forced to rely on data from three different samples of children. Children in the upper grades of elementary schools can complete only a limited number of loneliness measures within a single class period (the typical time allotted to psychological research by school principals). As a result of these constraints, a complete multitrait–multimethod approach, in which all children completed all six loneliness measures, was not feasible. An alternative approach was therefore adopted in which the three samples of children completed different combinations of the six measures. In each sample, then, a crucial part of the latent structure model could be tested, which comprised two or three of the latent constructs in the full model. To ensure comparability, the three studies were conducted at the same time, and children's ethnic background and socioeconomic status, as well as the consent rules and testing procedure adopted, were identical in the three cases.

Study 1

Method

Participants and Procedure

Sample 1 ($N = 292$) comprised students from five schools in the Dutch-speaking part of Belgium (Europe). All children attended the upper grades (Grade 5 or 6) of elementary school. The sample was composed of 148 fifth graders (64 girls and 84 boys) and 144 sixth graders (78 girls and 66 boys). Mean ages for the fifth- and sixth-grade groups were 10 years and 5 months ($SD = 4$ months) and 11 years and 5 months ($SD = 5$ months), respectively. All students were Caucasian and all schools were known to attract mainly students with a middle-class background.

All school principals were extensively briefed about the purpose of the study, and all granted permission to conduct the research in their schools after they had examined and approved the materials presented to the children. Under Belgian law, this is the only type of permission needed to conduct psychological research on school-age children. At the start of each testing session, it was made clear to the children that they could decline to take part in the study if they so desired. None of the children opted to do so. The questionnaires have been widely used and there is no evidence of risk associated with completing this type of instruments. All research assistants were trained to deal with any upset child and were specifically instructed to stop testing when it became too demanding for a child, to attend to the child's distress, and if necessary, to ensure adequate referral to appropriate services, in close collaboration with the school administrators.

All testing sessions took place in January 1998 during regularly scheduled class periods. All of the students in Grades 5 and 6 who were present on the day the research took place were asked to complete the questionnaires. The testing sessions were all supervised by a female undergraduate student in psychology who had received elaborate instructions for the administration of the measures from the first author. In all cases, one class was tested at a time and virtually all children managed to complete all measures within 50 min, that is, the typical duration of a class period in Belgium.

Measures

The children completed three loneliness measures in the following order: the LACA, the CLS, and the RPLQ.

LACA (Marcoen et al., 1987). This multidimensional measure of loneliness comprised 4 subscales of 12 items each (for a total of 48 items). These subscales concentrated on (a) loneliness in the relationships with parents (L–Parents), (b) loneliness in the relationships with peers (L–Peers), (c) aversion to aloneness (or A–Negative), and (d) affinity for aloneness (or A–Positive). Sample items for these scales were as follows: "My parents are ready to listen to me and to help me" (L–Parents; reverse scored); "I feel abandoned by my friends" (L–Peers); "I feel unhappy when I have to do things on my own" (A–Negative); and "At home I am looking for moments to be alone, so that I can do things on my own" (A–Positive), respectively. All items were answered on a 4-point scale, ranging from 1 (*never*) to 4 (*often*). Children's scores on the four subscales, therefore, could range between 12 and 48, with higher scores indicating higher degrees of loneliness and more negative or more positive attitudes toward being alone, respectively. (An English translation of all LACA items, which were originally developed in Dutch, can be found in the appendixes to Marcoen et al. 1987; Marcoen & Goossens, 1993; and Terrell-Deutsch, 1999.) Concurrent and construct validity of the measure, including associations between peer-related loneliness and peer rejection as assessed through sociometric techniques (Goossens, Scholte, & van Aken, 2000), was established in earlier research.

CLS (Asher & Wheeler, 1985). This was a 24-item measure that comprised 16 primary items, which were designed to tap children's feelings of loneliness in the school context. A sample item was "I get along with my classmates" (reverse-scored). The eight additional items were filler items that concentrated on children's hobbies and preferred activities and school subjects. An example of such a filler item was "I watch TV a lot." All items were answered on a 5-point scale ranging from 1 (*not at all*) to 5 (*always*). Participants' scores on the CLS, therefore, could range between 16 and 80, with higher scores indicating higher degrees of loneliness. Construct validity of the scale was demonstrated in several studies by means of associations with sociometric ratings by classmates (Asher, Parkhurst, Hymel, & Williams, 1990).

The CLS was initially developed in English and, therefore, was translated into Dutch, the native language of the children. In a first step of the translation process, the senior author and an undergraduate student in psychology each made an independent translation of the scale and agreed on a common version, following discussion. This preliminary version was then double-checked by a third person with a doctorate degree in psychology and an intimate knowledge of the target language (i.e., Dutch). In a second step, the CLS items were back-translated into English by a doctorate student in psychology, who had not been involved in the original translation process. To check the accuracy of the translation, another doctorate student in psychology matched the original English items and the items back-translated into English. Perfect matching was achieved and no difficulties arose during the procedure, which seemed to support the adequacy of the CLS translation used. In short, every effort was made to ensure that the concepts and phrases used in the CLS had similar meanings across cultural contexts (Behling & Law, 2000). Similar translation procedures were used with all the other loneliness scales originally developed in English that were used in this study.

RPLQ (Hayden-Thomson, 1989). This was a multidimensional measure of loneliness that comprised four subscales of 7 items each (for a total of 28 items). The items concentrated on two aspects of social satisfaction, that is, group integration and personal intimacy, as experienced within the peer group and the family, respectively. (These aspects of social satisfaction were referred to as "social provisions" by Weiss, 1974.) All items were reverse-scored and summed within each relationship to yield measures of two aspects of loneliness, that is, lack of integration and lack of intimacy.

Sample items for the four scales (all reverse-scored) were "I feel that I usually fit in with my family" (Lack of Family Integration); "I have someone in my family I can tell everything to" (Lack of Family Intimacy); "I feel in tune with other children" (Lack of Peer Integration); and "I have a friend I can tell everything to" (Lack of Peer Intimacy), respectively. Because each item was again answered on a 5-point scale ranging from 1 (*not at all*) to 5 (*always*), children's scores on each of the four subscales could range between 7 and 35. Higher scores again indicated higher levels of loneliness for each subscale. The scale was originally developed in Canada for use with English-speaking children (Hayden-Thomson, 1989), and its concurrent validity has been established in studies conducted in that country (McDougall & Hymel, 1998; Rubin, Chen, McDougall, Bowker, & McKinnon, 1995; Terrell-Deutsch, 1999).

Results and Discussion

Internal consistencies (Cronbach's α) and intercorrelations for all loneliness measures used with Sample 1 are presented in Table 1. Most scales and subscales evidenced high reliability (Cronbach's α .80 or above), with the exception of the Lack of Peer Intimacy subscale of the RPLQ and the two scales probing children's attitude toward being alone (as found in the LACA).

The intercorrelations among the measures used with Sample 1 allowed for a partial test of the latent structure model described in the introduction. Specifically, three subscales tapped parent-related (or family-related) loneliness (the first three scales in Table 1), whereas four scales might be considered measures of peer-related loneliness (Scales 4 through 7 in Table 1). Because aversion to aloneness and affinity for aloneness were each measured by a single subscale only, these two subscales (Scales 8 and 9 in Table 1) were dropped from the latent structure analysis. CFA (maximum likelihood method) was conducted on the covariance matrix using LISREL 8.30 (Jöreskog & Sörbom, 1993).

The two-factor model was compared to a one-factor model (with all seven scales loading on a general loneliness factor) and a slightly modified two-factor model. In the latter model, the error variances of the CLS and the LACA peer-related subscale were allowed to correlate. This modification undoubtedly reflected item overlap among these two measures. (The scales have one item in common—"I feel alone at school"—and several other LACA items are similar in content to CLS items.) Fit indexes for all three models are presented in Table 2 (upper half).

In that table, the chi-square value should be as small as possible, whereas the other fit indexes, which are less sensitive to sample size, can range between 0 and 1. Both the root mean square error of approximation and the standardized root mean square residual again should be as small as possible and smaller than .05, if possible. The Adjusted Goodness-of-Fit Index and the

Table 1. *Internal Consistency (Cronbach's α) and Intercorrelations Among the Loneliness Measures (Sample 1)*

Scale	2	3	4	5	6	7	8	9	Alpha	*M*	*SD*
1. L–Parents (LACA)	.69***	.59***	.18**	.24***	.15**	.23***	.03	.05	.81	17.87	5.04
2. Lack of Family Integration (RPLQ)		.70***	.19***	.34***	.29***	.22***	.05	.17**	.82	15.04	5.15
3. Lack of Family Intimacy (RPLQ)			.07	.17**	.27***	.09	.01	.12*	.80	14.81	5.50
4. L–Peers (LACA)				.61***	.38***	.73***	.16**	.33***	.86	23.32	6.83
5. Lack of Peer Integration (RPLQ)					.51***	.72***	−.10	.24***	.81	17.03	4.78
6. Lack of Peer Intimacy (RPLQ)						.47***	−.15*	.17**	.78	15.46	5.46
7. Children's Loneliness Scale (CLS)							−.04	.24***	.87	33.11	9.98
8. A–Negative (LACA)								.01	.74	33.53	5.79
9. A–Positive (LACA)								—	.79	30.50	6.13

Note: Confirmatory factor analyses were performed on the covariance rather than the correlation matrix. L = loneliness; A = aloneness. LACA = Loneliness and Aloneness Scale for Children and Adolescents; RPLQ = Relational Provisions Loneliness Questionnaire; CLS = Children's Loneliness Scale.
*$p < .05$. **$p < .01$. ***$p < .001$.

Table 2. *Fit Indexes for Various Structural Models of Loneliness*

Model Description		Fit Indexes					
Model	*df*	χ^2	RMSEA	AGFI	CFI	SRMR	AIC
Study 1							
One-factor	14	512.77	.38	.26	.50	.22	620.70
Two-factor	13	59.95	.12	.87	.95	.06	93.49
Modified two-factor	12	39.90	.09	.92	.97	.05	69.81
Study 2							
One-factor	14	213.37	.27	.53	.82	.11	286.16
Two-factor	13	65.23	.13	.85	.95	.06	94.51
Three-factor	11	83.37	.16	.78	.93	.06	112.80
Modified two-factor	10	27.72	.09	.91	.98	.03	63.87

Note: RMSEA = root mean square error of approximation; AGFI = Adjusted Goodness-of-Fit Index; CFI = comparative fit index; SRMR = standardized root mean square residual; AIC = Akaike's Information Criterion.

comparative fit index, by contrast, should be as large as possible and should exceed the value of .95, if possible. Akaike's Information Criterion, finally, takes parsimony as well as fit into account. This index should not increase when moving from less complex to more complex models (i.e., when the number of degrees of freedom decreases). It is clear from Table 2 that the third model (the modified two-factor model) provided the best fit to the data.

Comparison of the two-factor model (Model 2) with the one-factor model (Model 1) led to the rejection of the latter model ($\Delta\chi^2 = 452.82$, $df = 1$, $p < .001$). A slight modification to the two-factor model, as described earlier, further improved model fit. This final model (Model 3) is represented in Figure 1. The peer-related loneliness construct was mainly defined in terms of lack of group integration, with a lower standardized loading for the Lack of Peer Intimacy subscale of the RPLQ. Parent-related loneliness, by contrast, was defined in terms of both lack of group integration and lack of intimacy within the family. In short, considerable support was found for a two-construct model of childhood loneliness in which parent- and peer-related loneliness were distinguished. The two constructs were only modestly correlated.

Study 2

Method

Participants and Procedure

Sample 2 ($N = 241$) drew children from three schools from the Dutch-speaking part of Belgium. A breakdown by grade yielded 164 fifth graders (80 girls and 83 boys) and 77 sixth graders (64 girls and 14 boys). Mean ages for the fifth- and sixth-grade groups were 10 years and 6 months ($SD = 4.5$ months) and 11 years and 6 months ($SD = 5$ months), respectively.

Measures

The children completed four loneliness measures in the following order: LACA, CLS, CMLS, and the CELS. The first two of these instruments were also used with Sample 1 and were described previously.

CMLS (DeBiase, 1992). This instrument comprised three subscales of 10 items each: parent-related loneliness, peer-related loneliness, and global loneliness. Sample items for the three subscales were "My parents do not really listen to me when I talk to them" (Parent-Related Loneliness); "I feel left out when I'm with other kids my age" (Peer-Related Loneliness); and "I don't have anyone I feel close to" (Global Loneliness), respectively. The 5-point scale ranging from 1 (*not at all*) to 5 (*always*) was again used for this instrument and, as a result, participants' scores on the three scales could range between 10 and 50. Construct validity of the instrument was established in earlier research through significant associations between the Peer-Related Loneliness subscale and sociometric ratings. Concurrent validity was evidenced through correlations with social support, trait anxiety, and depression (DeBiase, 1992).

CELS (Heinlein & Spinner, 1986). This 11-item instrument was designed to tap feelings of loneliness that resulted from a sense of emotional isolation, that is, from a lack of close personal attachments (Weiss, 1973). Sample items were "I feel part of my family" (reverse-scored) and "I feel nobody cares for me anymore." The 5-point scale ranging from 1 (*not at all*) to 5 (*always*) was again used with all items of the measure, so that children's total score could range between

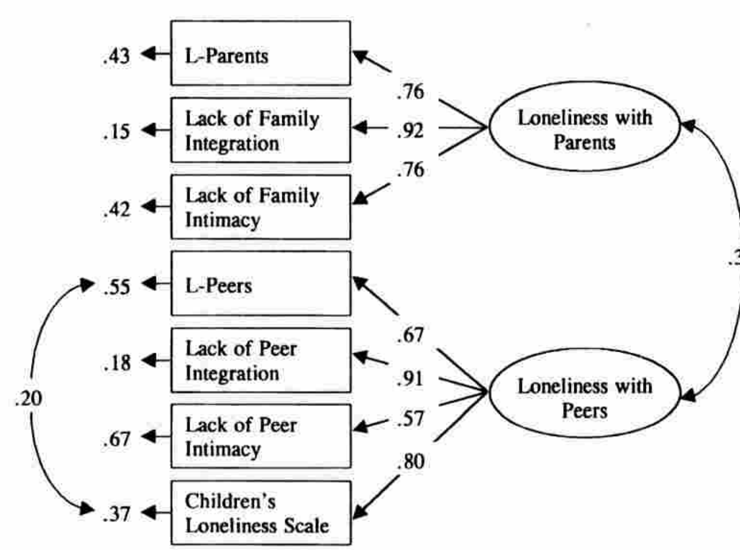

Figure 1. Standardized solution of the CFA for Study 1. All loadings and correlations were significant at $p < .01$.

11 and 55. Concurrent validity was demonstrated in earlier work by significant associations with measures of related constructs, such as perceived parenting style and the quality of children's relational network (Heinlein & Spinner, 1986).

Results and Discussion

Internal consistencies (Cronbach's α) and intercorrelations for all loneliness measures used with Sample 2 are presented in Table 3. Only two scales (the CLS and the Peer-Related Loneliness subscale of the LACA) evidenced high internal consistency (Cronbach's α .80 or above). The other scales were only moderately reliable (with αs in the .70s).

The intercorrelations among the measures used with Sample 2 allowed for a partial test of the latent structure model that was very similar to the test performed with Sample 1. Once again, three subscales were considered measures of parent-related (or family-related) loneliness (the first three scales in Table 3), whereas four scales tapped peer-related loneliness (Scales 4 through 7 in Table 3). Because aversion to aloneness and affinity for aloneness were again measured by a single subscale only, these two subscales (Scales 8 and 9 in Table 3) were dropped from the latent structure analysis.

The two-factor model was compared to a one-factor model, a slightly modified two-factor model, and a three-factor model. Two modifications were made to the initial two-factor model. First, the CELS was allowed to load on the Peer-Related Loneliness construct as well, whereas the Global Loneliness subscale of the CMLS was allowed a subsidiary loading on the Parent-Related Loneliness construct. Second, the error variances of the CLS and the Peer-Related Loneliness subscale of the LACA were again allowed to correlate, as was the case in Study 1. In the three-factor model, the first two scales in Table 3 loaded on a Parent-Related Loneliness factor, Scales 4, 5, and 7 defined a Peer-Related Loneliness factor, and Scales 3 and 6 (the CELS and the Global Loneliness subscale of the CMLS) loaded on a third factor that reflected loneliness with regard to the generalized other. Fit indexes for all four models are presented in Table 2 (lower half).

In terms of these fit indexes, Model 4 (the modified two-factor model) provided the best fit to the data. Comparisons of the two-factor model (Model 2) with the one-factor model (Model 1; $\Delta\chi^2 = 148.14$, $df = 1$, $p < .001$) and the three-factor model (Model 3; $\Delta\chi^2 = 18.14$, $df = 2$, $p < .001$) confirmed that Model 2 was superior to those two alternative models. Slight modifications to this model, as described earlier, further improved model fit. This final model (Model 4) is represented in Figure 2. By way of a conclusion, considerable support was again found for a two-construct model of childhood loneliness in which parent- and peer-related Loneliness were distinguished. Due to the subsidiary loadings of some of the measures, the correlation between the two constructs was somewhat higher than in Study 1, but was still in the moderate range.

Study 3

Method

Participants and Procedure

Sample 3 ($N = 225$) comprised children from five schools from the Dutch-speaking part of Belgium. This sample was composed of 110 fifth graders (67 girls and 43 boys) and 115 sixth graders (51 girls and 64 boys). Mean ages for the fifth and sixth grade groups were 10

Table 3. *Internal Consistency (Cronbach's α) and Intercorrelations Among the Loneliness Measures (Sample 2)*

Scale	2	3	4	5	6	7	8	9	Alpha	*M*	*SD*
1. L–Parents (LACA)	.65***	.59***	.30***	.34***	.44***	.43***	.07	.13*	.79	16.92	4.50
2. Parent-related Loneliness (CMLS)		.67***	.32***	.38***	.50***	.50***	.13*	.15*	.72	15.78	5.72
3. Emotional Loneliness (CELS)			.44***	.50***	.63***	.54***	.13*	.04	.70	18.51	5.27
4. L–Peers (LACA)				.72***	.64***	.77***	.27***	.33***	.89	22.91	7.50
5. Peer-related Loneliness (CMLS)					.74***	.77***	.16*	.21***	.78	20.58	6.25
6. Global Loneliness (CMLS)						.72***	.14*	.21***	.75	20.24	5.50
7. Children's Loneliness Scale (CLS)							.17**	.24***	.87	32.68	10.28
8. A–Negative (LACA)								.18**	.72	32.98	5.67
9. A–Positive (LACA)								—	.74	29.79	5.57

Note: Confirmatory factor analyses were performed on the covariance rather than the correlation matrix. L = loneliness; A = aloneness; LACA = Loneliness and Aloneness Scale for Children and Adolescents; CLS = Children's Loneliness Scale; CMLS = Children's Multidimensional Loneliness Scale; CELS = Children's Emotional Loneliness Scale.
*$p < .05$. **$p < .01$. ***$p < .001$.

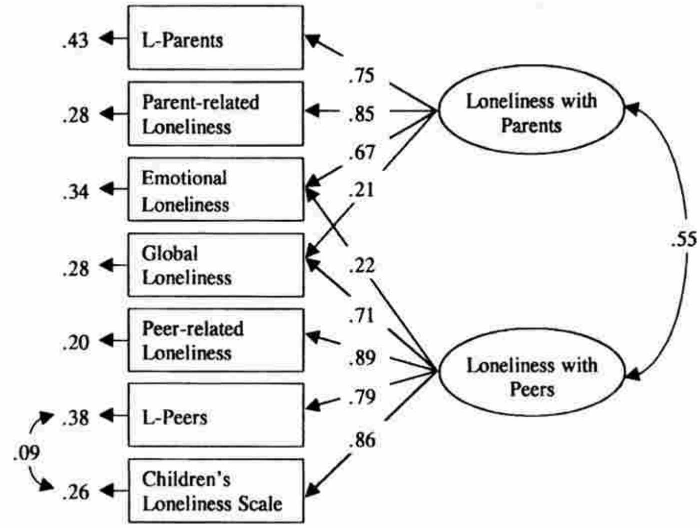

Figure 2. Standardized solution of the CFA for Study 2. All loadings and correlations were significant at $p < .01$.

years and 5 months (SD = 7 months) and 11 years and 3 months (SD = 6 months), respectively.

Measures

The children completed three loneliness measures in the following order: LACA, CLS, and ABAQ. The first two of these instruments were identical to the measures used with Samples 1 and 2 and were described previously.

ABAQ (Youngblade et al., 1999). This instrument included 37 items in question format that measured children's perceptions of being alone. Twelve of these items were filler items on hobbies and preferred activities (similar but not identical to the fillers in the CLS). An example of such an item is "Do you like to go to the movies?" The 25 primary items were split into two subscales that were labeled Aversion to Being Alone (12 items) and Ability to Be Alone (13 items), respectively. Sample items were "Do you feel sad when you have to spend time on your own?" for Aversion to Being Alone and "Do you like being by yourself?" for Ability to Be Alone. All items were answered using a 3-point scale using 2 (*yes*), 1 (*sometimes*), and 0 (*no*). As a consequence, children's scores on the two subscales could range between 0 and 24 (for the Aversion to Being Alone subscale) and 0 and 26 (for the Ability to Be Alone subscale), respectively. Concurrent and construct validity of the two subscales was established through associations with the CLS and with teachers' ratings of children's classroom behavior with peers (Youngblade et al., 1999).

Results and Discussion

Internal consistencies (Cronbach's α) and intercorrelations for all loneliness measures used with Sample 3 are presented in Table 4. The CLS and the Peer-Related Loneliness subscale of the LACA again proved to be highly reliable (Cronbach's α .80 and above), and the same held for the Aversion to Being Alone subscale of the LACA and the Ability to Be Alone subscale of the ABAQ. All other scales were moderately reliable (Cronbach's αs in the .70s). Means and standard deviations of the measures used with all three samples (LACA and CLS) were highly similar to the descriptive statistics obtained on Samples 1 and 2 (see Tables 1 and 3, respectively) and to estimates obtained in earlier work (Marcoen et al., 1987).

The intercorrelations among the measures used with Sample 3 allowed for a slightly different but again partial test of the latent structure model. Specifically, two subscales each were considered measures of peer-related loneliness (Scales 1 and 2 in Table 4), aversion to aloneness (Scales 3 and 4 in Table 4) and affinity for aloneness (Scales 5 and 6 in Table 4), respectively. Because parent-related loneliness was measured by a single subscale only, this subscale (Scale 7 in Table 4) was dropped from the latent structure analysis.

This three-construct model, however, could not be tested using CFA, due to identification problems (i.e., there was not enough information in the covariance matrix to estimate the model parameters properly; Kenny, Kashy, & Bolger, 1998). To avoid these difficulties, we conducted exploratory factor analysis. The correlations among the six indicators were factor-analyzed (principal axis factoring with iterations). Three factors were retained (based on the scree test and the eigenvalue > 1 rule) and rotated using the varimax criterion. Results of this exploratory factor analysis, which accounted for 66% of the variance, are presented in Table 5.

The two measures of negative attitudes toward being alone had substantial loadings on the first factor,

Table 4. *Internal Consistency (Cronbach's α) and Intercorrelations Among the Loneliness Measures (Sample 3)*

Scale	2	3	4	5	6	7	Alpha	*M*	*SD*
1. L–Peers (LACA)	.66***	.13*	.19*	.39***	.18*	.34***	.86	22.84	6.94
2. Children's Loneliness Scale (CLS)		–.09	.08	.13	.12	.35***	.81	33.03	10.03
3. A–Negative (LACA)			.53***	.07	–.35***	.07	.80	32.47	6.24
4. Aversion to Being Alone (ABAQ)				–.15*	–.52***	.03	.78	11.73	4.30
5. A–Positive (LACA)					.47***	.27***	.77	30.68	5.86
6. Ability to Be Alone (ABAQ)						.16**	.80	10.34	4.81
7. L–Parents (LACA)						—	.78	17.37	4.47

Note: L = loneliness; A = aloneness; LACA = Loneliness and Aloneness Scale for Children and Adolescents; ABAQ = Ability to Be Alone Questionnaire; CLS = Children's Loneliness Scale.
$^{*}p < .05$. $^{**}p < .01$. $^{***}p < .001$.

Table 5. *Factor Loadings Following Varimax Rotation for the Childhood Loneliness Measures Used With Sample 3*

Subscale	Factor I	Factor II	Factor III	h^2
L–Peers (LACA)	.18	**.81**	.34	.82
Children's Loneliness Scale (CLS)	–.07	**.82**	.00	.68
A–Negative (LACA)	**.75**	–.05	.10	.58
Aversion to Being Alone (ABAQ)	**.74**	.16	–.23	.64
A–Positive (LACA)	.00	.15	**.77**	.61
Ability to Be Alone (ABAQ)	**–.54**	.10	**.60**	.66

Note: Loadings higher than .35 are in bold.

the CLS and the Peer-Related Loneliness subscale of the LACA had a high loading on the second factor, and the two measures of positive attitudes toward aloneness defined a third factor. The substantial negative cross-loading of the Ability to Be Alone subscale (ABAQ) on the first factor seemed to indicate that children who state that they can manage time spent on their own do not have negative attitudes towards aloneness. Greater affinity for aloneness (LACA), by contrast, is not necessarily associated with less negative attitudes toward being alone.

Thus, some support was found, through exploratory factor analysis rather than CFA methods, for a three-construct model of childhood loneliness and related constructs. In this model, peer-related loneliness, aversion to aloneness, and affinity for aloneness were clearly distinguished from one another.

General Discussion

This study has significantly expanded the available knowledge base on the measurement of childhood loneliness. The internal consistency of an unusually large number of loneliness measures was examined in comparative fashion. CFA supported the conceptualization of parent- and peer-related loneliness as two different constructs (two-factor model) rather than alternative models that were both simpler (one-factor model) and more complex (three-factor model) in nature.

Internal Consistency

In terms of homogeneity, the CLS and the Peer-Related Loneliness subscale of the LACA were most reliable. Cronbach's α for these two scales exceeded .80 in all three samples. The other scales and subscales were somewhat less reliable (with αs typically ranging between .70 and .80). The least homogeneous measure was the CELS, which may reflect this scale's diverse content. Despite these remarks, all measures are sufficiently reliable to be used in correlational research.

Confirmatory and Exploratory Factor Analysis

Partial support was found for a latent structure model of childhood loneliness and related constructs in which four constructs are distinguished. In partially overlapping analyses, the 13 scales and derivative subscales of the six measures were shown to tap four different but interrelated latent constructs: peer-related loneliness, family-related loneliness (in many cases restricted to parent-related loneliness), aversion to being alone, and affinity for being alone. There were several indications that the three samples were highly comparable on a number of crucial variables, such as average age and average scores on the two loneliness measures completed by all participants. In future work, however, researchers should try to test the full, four-construct model on a single sample, to avoid issues of comparability across samples.

Recommendations for Use

Because all measures examined were at least moderately reliable, the selection among these scales will depend on the particular objectives that researchers and clinicians have in mind. If researchers want to concentrate on loneliness in children's relationships with their peers, use of the CLS or the peers subscale of the LACA, which have a lot in common, can be recommended. The peers subscale of the CMLS, the Peer Group Integration, and, to a lesser extent, the Peer Intimacy subscale of the RPLQ may be suitable alternatives for those two measures. Latent structure analysis indicates that the Global Loneliness subscale of the CMLS is not a good choice in this regard.

If loneliness as experienced within the family is the topic under investigation, then the Parent-Related Loneliness subscales found in the LACA and the CMLS or the family scales of the RPLQ can all be recommended. The CELS seems less suitable in this regard, due to its diverse item content. If, however, children's attitude to being alone is the focus of interest, then researchers and clinicians basically have two options. Both the LACA subscales on Aversion to and Affinity for Aloneness or the Aversion to and Ability for Aloneness scales of the ABAQ could be used for this purpose.

Finally, researchers and clinicians alike may want to explore children's loneliness across different relational contexts and to collect data on their attitude toward time spent alone. With this admittedly broad objective in mind, researchers and clinicians can assemble a special purpose packet of scales, such as the parents and peers subscales of the CMLS combined with the ABAQ, to probe both loneliness and solitude. However, use of the LACA is advised in that case, because that scale combines measures of all four constructs identified in this study (i.e., peer-related loneliness, parent-related loneliness, aversion to aloneness, and affinity for aloneness) into a single instrument.

Limitations and Suggestions for Future Research

This study has established substantial convergence among multiple measures of the same variable. However, this can be but a first step in the validation process of any instrument. The next step entails that researchers demonstrate that what the measures have in common is in fact the construct they are supposed to tap (Nunnally & Bernstein, 1994). In other words, they have to use criterion measures of childhood loneliness and related constructs and establish significant associations between those criteria and the loneliness scales. The fact that no such criteria were used in this study is a major limitation. As described in the Method section, earlier work has already demonstrated the construct validity of some of the peer-related loneliness scales through associations with sociometric ratings by classmates. Construct validity for the parent-related loneliness measures could be established through parents' ratings of their relationships with their children or by means of observational techniques. Consistent use of multiple informants would avoid the problem of shared method variance that plagues all research that exclusively relies on self-report measures.

Second, because this study was conducted on Dutch-speaking children in Belgium, it is unclear whether the findings can be generalized to children in North America (the United States and Canada), on whom the bulk of childhood loneliness research has been performed. The fact that English-language adaptations of the LACA, which was originally developed for use with Dutch-speaking children, have been used successfully and were highly reliable with both Canadian children (Terrell-Deutsch, 1999) and Irish adolescents (de Roiste, 2000) seems to be promising in this regard. Finally, the study focused on normal children who were functioning effectively in elementary school. Whether the findings can be extended to clinical groups of children is again an empirical matter to be determined in future research.

All three limitations lead directly to recommendations for future research. Researchers on both sides of the Atlantic can expand the available knowledge base on the measurement of childhood loneliness through replications of this study using different sets of measures as completed by normal and clinical groups of children with different ethnic backgrounds and through comparative analyses of construct validity using multiple informants (i.e., self- and other-reports). In terms of alternative measures, researchers may want to include childhood loneliness measures that have recently become available, such as the Peer Network and Dyadic Loneliness Scale (Hoza, Bukowski, & Beery, 2000) or a recent adaptation of the CLS, in which loneliness per se and social dissatisfaction are distinguished (Burgess, Ladd, Kochenderfer, Lambert, & Birch, 1999).

References

Asher, S. R., Hymel, S., & Renshaw, P. (1984). Loneliness in children. *Child Development, 55,* 1456–1464.

Asher, S. R., Parkhurst, J. T., Hymel, S., & Williams, G. A. (1990). Peer rejection and loneliness in childhood. In S. R. Asher & J. D. Coie (Eds.), *Peer rejection in childhood* (pp. 253–273). New York: Cambridge University Press.

Asher, S. R., & Wheeler, V. A. (1985). Children's loneliness: A comparison of rejected and neglected peer status. *Journal of Consulting and Clinical Psychology, 53,* 500–505.

Behling, O., & Law, K. S. (2000). *Translating questionnaires and other research instruments: Problems and solutions* (Sage University Paper Series on Quantitative Applications in the Social Sciences, No. 07-131). Thousand Oaks, CA: Sage.

Boivin, M., Hymel, S., & Bukowski, W. M. (1995). The roles of social withdrawal, peer rejection, and victimization by peers in predicting loneliness and depressed mood in childhood. *Development and Psychopathology, 7,* 765–785.

Bullock, J. R. (1993). Children's loneliness and their relationships with family and peers. *Family Relations, 42,* 46–49.

Burgess, K. B., Ladd, G. W., Kochenderfer, B. J., Lambert, S. F., & Birch, S. H. (1999). Loneliness during early childhood: The role of interpersonal behaviors and relationships. In K. J. Rotenberg & S. Hymel (Eds.), *Loneliness in childhood and adolescence* (pp. 109–134). New York: Cambridge University Press.

Coyne, J. C., Burchill, S. A. L., & Stiles, W. B. (1991). An interactional perspective on depression. In C. R. Snyder & D. R.

Forsyth (Eds.), *Handbook of social and clinical psychology: The health perspective* (pp. 327–349). New York: Pergamon.

Crick, N. R., & Ladd, G. W. (1993). Children's perceptions of their peer experiences: Attributions, loneliness, social anxiety, and social avoidance. *Developmental Psychology, 29,* 244–254.

DeBiase, M. E. I. (1992). Development of a multidimensional loneliness measure for middle childhood. *Dissertation Abstracts International, 52,* p. 6655B. (University Microfilms No. DA 92-13437).

de Roiste, A. (2000). Peer and parent-related loneliness in Irish adolescents. *Irish Journal of Psychology, 21,* 237–246.

Goossens, L., & Marcoen, A. (1999). Adolescent loneliness, self-reflection, and identity: From individual differences to developmental processes. In K. J. Rotenberg & S. Hymel (Eds.), *Loneliness in childhood and adolescence* (pp. 225–243). New York: Cambridge University Press.

Goossens, L., Scholte, R., & van Aken, M. (2000, July). Loneliness, depression, and personality: Associations with sociometric status and peer reputation in early adolescence. In B. Laursen & W. M. Bukowski (Chairs), *The self and peer relations.* Symposium conducted at the 16th Biennial Meeting of the International Society for the Study of Behavioral Development (ISSBD), Beijing, People's Republic of China.

Hayden-Thomson, L. K. (1989). *Children's loneliness.* Unpublished doctoral dissertation, University of Waterloo, Ontario, Canada.

Heinlein, L., & Spinner, B. (1986). *The relationship between children's attachment experiences and experiences of emotional loneliness.* Unpublished manuscript, University of New Brunswick, Fredericton, Canada.

Hoza, B., Bukowski, W. M., & Beery, S. (2000). Assessing peer network and dyadic loneliness. *Journal of Clinical Child Psychology, 29,* 119–128.

Jöreskog, K. G., & Sörbom, D. (1993). *LISREL VIII users' reference guide.* Mooresville, IN: Scientific Software.

Joiner, T. E., Jr., Catanzaro, S. J., Rudd, M. D., & Rajab, M. H. (1999). The case for a hierarchical, oblique, and bidimensional structure of loneliness. *Journal of Social and Clinical Psychology, 18,* 47–75.

Kenny, D. A., Kashy, D. A., & Bolger, N. (1998). Data analysis in social psychology. In D. T. Hilbert, S. T. Fiske, & G. Lindzey (Eds.), *The handbook of social psychology* (4th ed., Vol. 1, pp. 233–265). New York: Oxford University Press.

Larson, R. W. (1990). The solitary side of life: An examination of the time people spend alone from childhood to old age. *Developmental Review, 10,* 155–183.

Long, J. S. (1983). *Confirmatory factor analysis: A preface to LISREL* (Sage University Paper Series on Quantitative Applications in the Social Sciences, No. 07-33). Beverly Hills, CA: Sage.

Marcoen, A., & Goossens, L. (1993). Loneliness, attitude towards aloneness, and solitude: Age differences and developmental significance during adolescence. In S. Jackson & H. Rodriguez-Tomé (Eds.), *Adolescence and its social worlds* (pp. 197–227). Hillsdale, NJ: Lawrence Erlbaum Associates, Inc.

Marcoen, A., Goossens, L., & Caes, P. (1987). Loneliness in pre- through late adolescence: Exploring the contributions of a multidimensional approach. *Journal of Youth and Adolescence, 16,* 561–577.

McDougall, P., & Hymel, S. (1998). Moving into middle school: Individual differences in the transition experience. *Canadian Journal of Behavioral Science, 30,* 108–120.

Nunnally, J. C., & Bernstein, I. H. (1994). *Psychometric theory* (3rd ed.). New York: McGraw-Hill.

Parker, J. G., & Asher, S. R. (1993). Friendships and friendship quality in middle childhood: Links with peer group acceptance and feelings of loneliness and social dissatisfaction. *Developmental Psychology, 29,* 611–621.

Parkhurst, J. T., & Asher, S. R. (1992). Peer rejection in middle school: Subgroup differences in behavior, loneliness, and interpersonal concerns. *Developmental Psychology, 28,* 231–241.

Peplau, L. A., & Perlman, D. (Eds.). (1982). *Loneliness: A sourcebook of current theory, research and therapy.* New York: Wiley.

Rotenberg, K. J. (1999). Childhood and adolescent loneliness: An introduction. In K. J. Rotenberg & S. Hymel (Eds.), *Loneliness in childhood and adolescence* (pp. 3–8). New York: Cambridge University Press.

Rubin, K. H., Chen, X., McDougall, P., Bowker, A., & McKinnon, J. (1995). The Waterloo Longitudinal Project: Predicting internalizing and externalizing problems in adolescence. *Development and Psychopathology, 7,* 751–764.

Terrell-Deutsch, B. (1990, May). *Assessing loneliness in children.* Paper presented at the biennial University of Waterloo Conference on Child Development, Waterloo, Ontario, Canada.

Terrell-Deutsch, B. (1999). The conceptualization and measurement of childhood loneliness. In K. J. Rotenberg & S. Hymel (Eds.), *Loneliness in childhood and adolescence* (pp. 11–33). New York: Cambridge University Press.

Weiss, R. S. (1973). *Loneliness: The experience of social and emotional isolation.* Cambridge, MA: MIT Press.

Weiss, R. S. (1974). The provisions of social relationships. In Z. Rubin (Ed.), *Doing unto others* (pp. 17–26). Englewood Cliffs, NJ: Prentice Hall.

Youngblade, L. M., Berlin, L. J., & Belsky, J. (1999). Connections among loneliness, the ability to be alone, and peer relationships in young children. In K. J. Rotenberg & S. Hymel (Eds.), *Loneliness in childhood and adolescence* (pp. 135–152). New York: Cambridge University Press.

Received January 30, 2001
Accepted November 13, 2001